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CLINICAL INVESTIGATION

Brachytherapy With 15- Versus 20-mm Ruthenium 106 Plaques Without Verification of Plaque Position Is Associated With Local Tumor Recurrence and Death in Posterior Uveal Melanoma

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Purpose: Brachytherapy with episcleral plaques is the most common primary tumor treatment for uveal melanoma. This study aimed to compare the risk of tumor recurrence and metastatic death between 2 frequently used ruthenium 106 plaque designs: CCB (20.2 mm) and CCA (15.3 mm).

Methods and Materials: Data were obtained from 1387 consecutive patients treated at St. Erik Eye Hospital, Stockholm, Sweden between 1981 and 2022 (439 with CCA and 948 with CCB plaques). During the period, scleral transillumination was performed to delineate tumor margins before plaque insertion, but accurate plaque positioning was not verified after scleral attachment, and no minimum scleral dose was used.

Results: Patients treated with CCA plaques had smaller tumors than those treated with CCB plaques (mean diameter, 8.6 vs 10.5 mm; $P < .001$). There were no differences in patient sex, age, tumor distance to the optic disc, tumor apex dose, dose rate, or in rates of ciliary body involvement, eccentric plaque placement, or adjunct transpupillary thermotherapy (TTT). The average difference between plaque and tumor diameter was greater with the CCB plaque, and a smaller difference was an independent predictor of tumor recurrence. The 15-year incidence of tumor recurrence was 28% and 15% after treatment with CCA and CCB plaques, respectively (competing risk analysis, $P < .001$). Multivariate Cox regression analysis revealed a lower risk for tumor recurrence with CCB plaques (hazard ratio, 0.50). Similarly, patients treated with CCB plaques had a lower risk for uveal melanoma−related mortality (hazard ratio, 0.77). The risk for either outcome was not lower for patients treated with adjunct TTT. Uni- and multivariate time-dependent Cox regressions demonstrated that tumor recurrence was associated with uveal melanoma−related and all-cause mortality.

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All data for this study are available at https://rcsyd.se. Approval from the Swedish Ethical Review Authority is required before access is granted.

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Conclusions: Compared with 20-mm plaques, brachytherapy with 15-mm ruthenium plaques is associated with a higher risk for tumor recurrence and death. These adverse outcomes may be avoided by increasing safety margins and implementing effective methods to verify accurate plaque positioning.

Introduction

Uveal melanoma has a high risk for metastatic progression and a 20-year relative survival rate of 60%.[1,2] Brachytherapy with radioactive episcleral plaques is the first-line treatment for most primary tumors.[3-5] Although several radioisotopes are available, γ-emitting iodine 125 and β-emitting ruthenium 106 plaques are widely adopted worldwide.[6-13] Unlike the γ radiation emitted by iodine 125 plaques, the β radiation emitted by ruthenium 106 has limited range and is therefore indicated only for tumors with an apical thickness of up to 6 mm.[14,15] The 2 most frequently used ruthenium 106 designs are the circular 20.2-mm CCB and 15.3-mm CCA plaques.[16-18] Within clinically relevant ranges, variations in the prescribed radioactive dose and dose rate are not strong predictors for local treatment failure or patient survival.[19,20] Suboptimal placement of the plaque in relation to the tumor in the lateral dimension or by tilting of the plaque away from the sclera might, however, lead to undertreatment of some tumor areas.[21-24] To ensure that an adequate radioactive dose is delivered to all parts of a tumor, a safety margin of at least 2 mm is therefore applied. Consequently, CCA and CCB plaques are typically used for lesions with a largest basal tumor diameter (LBD) of up to 11 and 16 mm, respectively.[25]

For small tumors, smaller plaque diameters have the benefit of irradiating a smaller volume of healthy tissue, which may help minimize collateral damage and reduce the risk of side effects such as radiation retinopathy and optic neuropathy. On the other hand, narrow safety margins could increase the risk for local treatment failure.[26] Many ocular oncologists use eccentric plaque positioning with the posterior edge aligned with the posterior tumor margin (ie, a 0-mm safety margin) to reduce radioactive doses to the optic nerve and fovea and thereby the risk of visual loss. This will reduce the radioactive dose to some areas of the tumor, all other things being equal, and has been associated with local tumor recurrence in a recent study by Jouhi et al.[16,26] Other studies, however, have found low local recurrence rates with CCA plaques and/or eccentric placement, when accurate plaque positioning has been ensured with special measures.[25,27]

Local recurrences are associated with significant distress to patients and require substantial resources from the health care system. Previous research has shown that patients prefer treatment alternatives with less risk for cancer recurrence over treatments with fewer side effects.[28] Moreover, tumor recurrence after primary treatment of uveal melanoma has been associated with an increased risk for mortality.[29-31]

In this study, our aim was to test the hypothesis that ruthenium plaque brachytherapy with smaller-diameter CCA plaques is not associated with a greater risk for tumor recurrence and mortality. To this end, we analyzed a large retrospective cohort of 1387 patients treated with CCA or CCB plaques over a span of 40 years. All treatments were performed at Sweden's only ocular oncology center, which gave our study a meticulous level of detail on baseline characteristics, treatment factors, and follow-up. We found that treatment with a smaller-diameter CCA plaque is associated with a greater risk for tumor recurrence and metastatic death than treatment with the larger CCB plaque.

Methods and Materials

Patients and samples

All 1387 patients treated for posterior uveal melanoma (involving the choroid and/or ciliary body) using a ruthenium CCA or CCB plaque (diameter 15.3 and 20.2 mm, respectively, both with a spherical radius of 12 mm, Fig. E1) at St. Erik Eye Hospital, Stockholm, Sweden between January 1, 1981, and October 31, 2022, were included in the study. Patients with iris melanoma and those treated with iodine plaques or other ruthenium plaque designs (eg, CIA, COB, or CIB) were excluded. Of the 1387 patients, 439 (32%) and 948 (68%) had been treated with CCA and CCB plaques, respectively. Our post hoc power calculation indicated that this sample would allow us to detect differences of approximately 5 percentage points, assuming an incidence of tumor recurrence of 10% in the CCB group based on a previous study of recurrence rates after plaque brachytherapy (α = 0.05; β = 0.80).[29] Assuming an incidence of mortality of 35%, as found in a previous study from our institution, the sample size would allow us to detect differences in mortality of approximately 8 percentage points.[32] Throughout the period, St. Erik Eye Hospital was the only institution in Sweden to diagnose and treat uveal melanoma with plaque brachytherapy, ensuring full capture of all such patients in the country. All plaques were delivered by Eckert & Ziegler BEBIG (Berlin, Germany), and source specification data from the manufacturer were independently verified by medical physicists at the Karolinska University Hospital, Stockholm. All primary tumors were diagnosed by ocular oncologists by slit-lamp biomicroscopy, indirect ophthalmoscopy, A- and B-scan ultrasonography, fundus imaging, optical coherence tomography, and fluorescein- or indocyanine green angiography, as needed. Transvitreal or transscleral biopsies were performed if a diagnosis could not be established from the clinical examination.

Plaque brachytherapy with ruthenium 106 plaques was used for tumors with an apical thickness of up to 5 to 6 mm, with larger tumors up to 10 mm treated with iodine 125 plaque brachytherapy and thicker tumors typically treated by enucleation. CCA and CCB plaques were recommended for tumors with an LBD of up to 11.3 mm and up to 16.2 mm, respectively. The prescription point was the tumor's thickness plus 1 mm for the sclera. One hundred Gy was prescribed to the tumor apex, but no minimum scleral dose was used. Extrascleral extension was an indication for enucleation, unless minimal. Tumor proximity to the optic disc (eg, <2 mm or similar) was not an absolute contraindication for brachytherapy, with 95 tumors touching the optic disc being treated with ruthenium 106 between 1984 and 2015, as reported previously.[33] For tumors touching the optic disc, transpupillary thermotherapy (TTT) was added to about half of all brachytherapy treatments.[33]

Over the study period, more than a dozen subspeciality-trained ocular oncologists had performed the plaque insertions. The procedure was performed under general anesthesia. To position the plaque in the anteroposterior dimension, an ink mark was placed on the sclera based on the plaque diameter and an assumed distance from the fovea to limbus of 30 mm. For example, if the posterior margin of the tumor was at a distance of 10 mm from the fovea, the ink mark for the anterior edge of the plaque would be placed 0 + 2 mm (safety margin) posterior to the limbus for a 20-mm plaque. In most cases, the sclera was also transilluminated to delineate the tumor, but if the transillumination failed to reveal the tumor margins clearly or if transillumination was deemed unnecessary, the ocular oncologist was guided only by narrow or wide field fundus photos for positioning of the plaque in the correct meridian. Extraocular muscles were removed if deemed appropriate. The inferior oblique muscle was not always removed when plaques were placed inferotemporally, which typically results in an increased distance between the plaque and the tumor. However, this increased distance was not taken into account during treatment planning or after plaque placement. For the last 2 decades, a portable ultrasound has been available in the operating room, which enables intraoperative verification of plaque-tumor alignment. In practice, however, it has been used very rarely (no data on the frequency of use available. Estimation is less than once per 100 procedures). Consequently, the practices did not undergo any significant changes during the period. Most patients were treated with plaque brachytherapy within 1 to 6 weeks from diagnosis (mean 28 days; standard deviation [SD], 12 days).

The following data were collected: sex; age at treatment; tumor diameter, thickness, and location; American Joint Committee on Cancer (AJCC) T-category (8th ed); AJCC stage; and long-term outcomes, which included tumor recurrence, secondary treatment, metastasis status, and cause of death.[34] These data were available in our digitalized treatment registry. The cause of death was fed to our registry from the National Cause of Death Register, which is based on medical death certificates that must be submitted to the National Board of Health and Welfare within 3 weeks of death. Death certificates were usually completed by a family physician, the physician last seeing the patient before death, or a forensic pathologist.[35] The study adhered to the tenets of the Declaration of Helsinki, and approval was obtained from the Swedish Ethical Review Authority (reference 2022-05357-01). Informed consent was waived because this was a retrospective chart review that did not influence patient management (ie, treatment, testing, follow-up, or information to patients), because it did not require collection of new sensitive information including patient names, personal identity numbers, addresses or other contact details, and because it did not include analyses of biologic tissues.

Follow-up

After treatment, patients underwent routine ophthalmologic assessment at 1, 3, 6, and 12 months. If local control had been achieved, the patient was then seen semiannually or annually until death. At each visit, the eye was examined with slit lamp biomicroscopy, ultrasonography, and fundus photography. If tumor regression was insufficient or the tumor progressed (to any degree, as determined by biomicroscopy and by measurable growth in fundus photographs and/or ultrasound scans) 6 months after brachytherapy or at any later point, secondary brachytherapy, enucleation, or TTT was recommended. Within a month of diagnosis, patients also underwent radiologic staging by chest x-ray and liver ultrasonography or by computed tomography of the chest and abdomen. Screening for liver metastases was repeated by 6-monthly ultrasonography or computed tomography for at least 5 years, and then when prompted by symptoms, palpable masses, deteriorating health, jaundice, weight loss, abnormal blood tests, or other features suggestive of metastasis.

Statistical methods

Differences with a $P < .05$ were considered significant, and all P values were 2-sided. The Shapiro-Wilk test was used to evaluate the deviation of continuous variables from a normal distribution. If the test was significant ($P < .05$), the Mann-Whitney U test was used to compare groups; otherwise, Student t test was used. Pearson's χ^2 test (if all fields had a sample of ≥5) or Fisher's exact test (if any field had a sample of <5) was used in contingency tables. The administered radioactive dose was calculated based on the independently verified dose rates and deviations from the planned treatment duration, as described previously[20]:

$$\frac{actual\ brachytherapy\ duration\ (eg,\ 36.50\ hours)}{planned\ brachytherapy\ duration\ (eg,\ 34.75\ hours)} \times prescribed\ dose = administered\ dose$$

Tumor recurrence was defined as tumor growth after primary plaque brachytherapy, with or without an initial period of inactivity or regression, that required secondary plaque brachytherapy, secondary enucleation, TTT, or other interventions. Five-year actuarial (life table) local recurrence-free survival was calculated for each treatment year, and the trend was analyzed with linear regression. Plaque diameter (20.2 or 15.3 mm) minus LBD was calculated for each tumor. This should not be confused with safety margin, which is the intended minimum tumor-free margin of plaque overlap outside the tumor in all directions. Cox regression hazard ratios (HRs) with 95% CIs for treatment failure and mortality were calculated. All significant predictors from univariate Cox regressions were included in multivariate analyses, excluding directly dependent factors. Additionally, all variables that differed with a $P < .10$ in descriptive statistics of patients treated with CCA versus CCB plaques were included as covariates in separate multivariate regressions. Patient age can be an important covariate, as age differences may bias the risk of dying between groups. However, the risk is not necessarily linear, and in partial residual plots (Schoenfeld residuals proportional hazards test) the distribution of patient age at diagnosis (in years and in decades) and natural log-transformed age were all nonrandom in relation to both uveal melanoma−related and all-cause mortality (Fig. E2). In Cox regression analyses, we therefore included patient age >70 versus ≤70 years as a categorical variable, based on a previous study where this threshold was shown to identify patients with uveal melanoma with poor prognosis.[36] Further, we used 2 complementary methods to determine whether our follow-up data after CCA versus CCB treatment met the proportional hazard assumption: First, we built a Cox regression model to calculate the HR for tumor recurrence, uveal melanoma −related mortality, and all-cause mortality with a time-dependent (the time variable $T \times$ [CCA vs CCB plaques]) versus a time-independent variable (CCA vs CCB plaques) as covariates. Second, we used a graphical approach and assessed log-minus-log-transformed survival curves for time to tumor recurrence and patient mortality. If P was >.05 in Cox regression with a time-dependent versus a time-independent variable and if survival curves were parallel without any crossing or divergence, the proportional hazard assumption was considered fulfilled. In the examination of the prognostic importance of tumor recurrence for uveal melanoma−related and all-cause mortality, tumor recurrence could not be analyzed as a normal covariate: it is an unknown factor at baseline and may have varying prognostic implications depending on when it occurs. Therefore, we analyzed tumor recurrences as a time-varying covariate, modeled as the time variable $T >$ (time to recurrence, or time > max follow-up time for patients not suffering from recurrence). Cumulative incidence function estimates from competing risk data were plotted with the cmprsk package for R, and the equality of survival distributions was tested with Gray's test for equality.[37] All statistical analyses except competing risk survival analyses were performed using IBM SPSS statistics version 27 (Armonk, NY) and GraphPad Prism version 9.3.0 (San Diego, CA).

Results

Descriptive statistics

Patients treated with CCA plaques had smaller tumors with less advanced AJCC T-categories and stages, but there were no significant differences in sex, age at diagnosis, tumor distance to the optic disc, tumor apex dose, dose rate, or rates of ciliary body involvement, eccentric plaque placement, or adjunct TTT between patients treated with CCA and CCB plaques (Table 1, Fig. 1A).

Nineteen of 439 patients treated with CCA plaques (4%) had tumors with LBD > 11 mm, and 28 of 948 patients treated with CCB plaques (3%) had tumors with LBD > 16 mm (thus having a plaque diameter minus LBD of 4 mm or less). The difference was not significant (χ^2, $P = .27$). However, the average difference between plaque diameter and LBD was significantly greater in the CCB group (mean, 9.7 mm; SD, 3.3) than in the CCA group (mean, 6.7 mm; SD, 2.4; Mann-Whitney U, $P <$ .001; Fig. 1B). The proportion of patients treated with adjunct TTT was 7% for CCA and 9% for CCB, and the proportion of patients treated without a safety margin (eccentric plaque placement) was 29% for CCA and 24% for CCB, respectively. The 5-year actuarial local recurrence-free survival increased with approximately 1% per year throughout the study period, indicating a learning curve (linear regression slope coefficient, 0.01; R^2, 0.56; $P < .001$; Fig. 1C). However, the year of treatment was not an independent predictor of metastatic death in multivariate Cox regression analysis, with plaque type and LBD as covariates (HR, 0.99; 95% CI, 0.97-1.00; $P = .11$).

The median survival time was 4.2 years (interquartile range [IQR], 4.1) in 310 patients who died of metastatic uveal melanoma and 9.8 years (IQR, 10.0) in patients who died of other causes. In 664 patients who were alive at the close of the study, the median follow-up was 10.8 years (IQR, 11.0). Our data for time to tumor recurrence and patient mortality met the proportional hazards assumption for Cox regression with a time-dependent versus a time-independent variable ($P > .16$). The visual assessment of log-minus-log-transformed survival curves agreed (Fig. E3).

Risk for tumor recurrence

In cumulative incidence function estimates with all-cause mortality as the competing event, the 5-, 10-, and 15-year incidence rates of tumor recurrence were 21%, 26%, and 28%, respectively, after primary brachytherapy with CCA plaques and 13%, 15%, and 15%, respectively, after treatment with CCB plaques ($P < .001$; Fig. 1D).

Table 1 Characteristics of included patients, tumors, and treatment variables

Characteristic	All patients, n = 1387	Treated with CCA plaques, n = 439	Treated with CCB plaques, n = 948	*P* value
Sex, no. (%)				.19
Female	694 (50)	231 (53)	463 (49)	
Male	693 (50)	208 (47)	485 (51)	
Age at diagnosis, mean (SD)	64 (14)	63 (13)	64 (14)	.43
Age at diagnosis, no. (%)				.06
>70 y	889 (64)	297 (68)	592 (62)	
≤70 y	498 (36)	142 (32)	356 (38)	
Tumor LBD, mean mm (SD)	9.9 (3.2)	8.6 (2.4)	10.5 (3.3)	<.001*
Tumor thickness, mean mm (SD)	4.3 (1.7)	4.1 (1.5)	4.4 (1.8)	.02*
Tumor distance to OD, mean mm (SD)	3.7 (2.9)	3.8 (2.9)	3.7 (3.0)	.25
AJCC T-category, no. (%)				<.001*
T1a	613 (44)	255 (58)	358 (38)	
T1b	13 (1)	6 (1)	7 (1)	
T2a	632 (46)	165 (38)	467 (49)	
T2b	17 (1)	5 (1)	12 (1)	
T2c	2 (<1)	0 (0)	2 (<1)	
T3a	94 (7)	8 (2)	86 (9)	
T3b	7 (1)	0 (0)	7 (1)	
T4a	9 (1)	0 (0)	9 (1)	
AJCC stage, no. (%)				<.001*
I	613 (44)	255 (58)	358 (38)	
IIA	645 (46)	170 (39)	474 (50)	
IIB	112 (8)	13 (3)	99 (10)	
IIIA	18 (1)	0 (0)	18 (2)	
Plaque diameter minus LBD, mean mm (SD)[†]	8.7 (3.3)	6.7 (2.4)	9.7 (3.3)	<.001*
No safety margin,[‡] no. (%)	355 (26)	125 (29)	230 (24)	.10
Planned dose to tumor apex, mean Gy (SD)	98.3 (11.8)	97.6 (10.7)	98.6 (12.3)	.15
Administered dose to tumor apex, mean Gy (SD)	98.4 (10.2)	97.4 (10.5)	98.7 (10.1)	.77
Dose rate at tumor apex, mean Gy/h (SD)	1.4 (0.8)	1.4 (0.8)	1.4 (0.7)	.46
Scleral dose, mean Gy (SD)	571 (338)	542 (360)	583 (327)	<.001*
Brachytherapy combined with TTT,[§] no. (%)	114 (8)	29 (7)	85 (9)	.14
Tumor recurrence, no. (%)	261 (19)	127 (29)	134 (14)	
Secondary treatment, no. (%)[‖]				
Plaque brachytherapy	102 (7)	57 (13)	45 (5)	
Enucleation	170 (12)	81 (19)	89 (9)	
TTT	27 (2)	16 (4)	11 (1)	

Abbreviations: AJCC = American Joint Committee on Cancer; LBD = largest basal tumor diameter; OD = optic disc; SD = standard deviation; TTT = transpupillary thermotherapy.

* Significant P values.
† Difference in plaque diameter minus LBD, for example, 5 mm when a 20.2-mm CCB plaque is used for a tumor with a diameter of 15.2 mm in its largest dimension.
‡ Intended minimal distance between plaque edge and tumor edge 0 mm (eccentric placement).
§ TTT to central tumor margins (ie, closest to the optic disc and fovea) at the time of primary brachytherapy.
‖ A patient could be given more than 1 type of secondary treatment (eg, TTT plus plaque brachytherapy), which is why the total number of secondary treatments was higher than the number of tumor recurrences.

In univariate Cox regressions, tumor recurrence was associated with greater tumor diameter (P = .002) and thickness (P < .001), absence of safety margin (eccentric placement, P < .001), smaller difference between plaque diameter and LBD (P < .001), increasing AJCC T-category (P < .001), increasing stage (P < .001), and CCA plaques (P < .001).

The prognostic significance of plaque type used was retained in multivariate analyses, with a 50% lower risk of

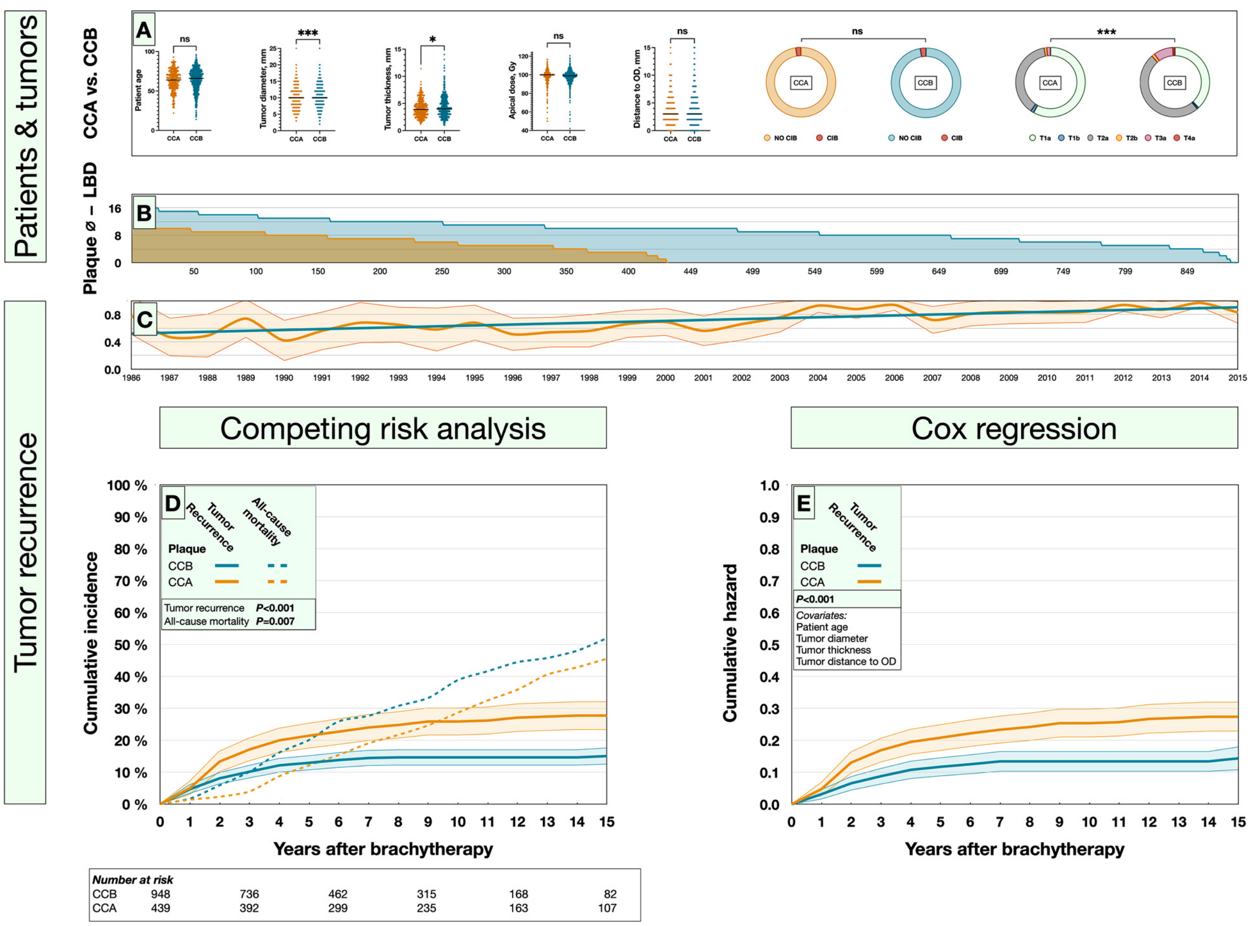

Fig. 1. Patient, tumor, and treatment characteristics versus risk for tumor recurrence. (A) Tumors treated with CCB plaques had significantly greater diameter, apical thickness, and American Joint Committee on Cancer T-category (categories <1% not shown). (B) Plaque diameter minus LBD for all tumors treated with CCA (brown) and CCB plaques (blue). The average difference was significantly greater in the CCB group (mean, 9.7 mm; standard deviation, 3.3) than in the CCA group (mean, 6.7 mm; standard deviation, 2.4; Mann-Whitney U, P < .001). (C) Five-year actuarial local recurrence-free proportion of patients treated each year between 1986 and 2015. Patients treated before 1986 (data available for <10 patients per year) and after 2015 (follow-up too short) not shown. In linear regression (blue line), the survival proportion increased with approximately 1% per year, indicating a learning curve. (D) Patients treated with CCA plaques had a greater cumulative incidence of tumor recurrence from competing risk data. (E) Similarly, patients treated with CCA plaques had a greater cumulative hazard for uveal melanoma–related mortality, adjusted for patient age, tumor diameter, tumor thickness, and tumor distance to the optic disc. Colored fields represent 95% CIs. *P < .05; ***P < .001. *Abbreviations*: LBD = largest basal tumor diameter; plaque ⌀ – LBD = plaque diameter minus LBD.

Table 2 Cox regression, hazard for tumor recurrence

Variate	B	S.E.	Wald	*P*	Exp(B)	95% CI lower	95% CI upper
Univariate							
Male sex	0.12	0.12	0.99	.32	1.13	0.89	1.44
Age at diagnosis*	−0.17	0.14	1.58	.21	0.84	0.64	1.10
LBD, mm†	0.06	0.02	9.30	.002**	1.06	1.02	1.10
Tumor thickness, mm†	0.14	0.03	19.09	<.001**	1.15	1.08	1.23
No safety margin‡	0.45	0.13	11.89	<.001**	1.56	1.21	2.02
Plaque diameter minus LBD, mm§	−0.12	0.02	34.52	<.001**	0.89	0.86	0.93
AJCC T-category‖	0.40	0.09	18.80	<.001**	1.49	1.24	1.78
AJCC stage‖	0.37	0.09	17.36	<.001**	1.44	1.21	1.71
Tumor distance to OD†	−0.04	0.02	2.78	.10	0.96	0.92	1.01
CIB	−0.18	0.45	0.15	.70	0.84	0.35	2.03
Tumor apex dose¶	−0.01	0.01	2.49	.11	0.99	0.98	1.00
Scleral dose¶	−0.00	0.00	0.38	.54	1.00	1.00	1.00
Brachytherapy combined with TTT#	−0.21	0.34	0.39	.53	0.81	0.41	1.58
CCB vs CCA plaque	−0.53	0.13	17.94	<.001**	0.59	0.46	0.75
Multivariate							
Age at diagnosis*	−0.17	0.15	1.27	.26	0.85	0.63	1.13
LBD, mm†	0.08	0.03	10.39	.001**	1.09	1.03	1.14
Tumor thickness, mm†	0.10	0.04	6.54	.01**	1.11	1.02	1.20
Tumor distance to OD†	−0.07	0.03	7.70	.006**	0.93	0.88	0.98
CCB vs CCA plaque	−0.70	0.15	22.87	<.001**	0.50	0.37	0.66
Multivariate							
No safety margin‡	0.23	0.17	1.87	.17	1.26	0.91	1.74
CCB vs CCA plaque	−0.68	0.16	17.72	<.001**	0.51	0.37	0.70
Multivariate							
No safety margin‡	0.50	0.13	14.58	<.001**	1.65	1.28	2.13
Plaque diameter minus LBD, mm§	−0.12	0.02	37.26	<.001**	0.89	0.85	0.92

Abbreviations: AJCC = American Joint Committee on Cancer; B = beta coefficient; CI = confidence interval; CIB = ciliary body involvement; LBD = largest basal tumor diameter; OD = optic disc; S.E. = standard error; TTT = transpupillary thermotherapy.

* >70 vs ≤70 years.

† Per millimeter.

‡ Intended minimal distance between plaque edge and tumor edge 0 mm (eccentric placement) vs ≥2 mm.

§ Per millimeter difference in plaque minus tumor diameter. Automatically dropped out of multivariate analyses due to collinearity with tumor diameter.

‖ For each step in AJCC T-category 1 to 4/stage I to IIIA.

¶ Per Gy.

\# TTT to central tumor margins (ie, closest to the OD and fovea) at the time of primary brachytherapy.

** Significant *P* values.

recurrences after CCB brachytherapy (HR, 0.50; 95% CI, 0.37-0.66; Fig. 1E, Table 2).

Risk for mortality

As expected, in cumulative incidence function estimates with death from any other cause as the competing event, the incidence of uveal melanoma−related mortality increased with AJCC stage, with a 10-year incidence of 13% for patients in stage I and 47% for patients in stage IIIA (Fig. E4).

The incidence of uveal melanoma−related mortality was similar after plaque brachytherapy with CCA and CCB plaques. Patients treated with CCB plaques had a slightly higher incidence of death from other causes (34% vs 28% at 15 years; $P = .003$; Fig. 2A).

In univariate Cox regression, CCB plaques were not associated with a greater risk for uveal melanoma−related mortality, as could have been expected by their use for larger

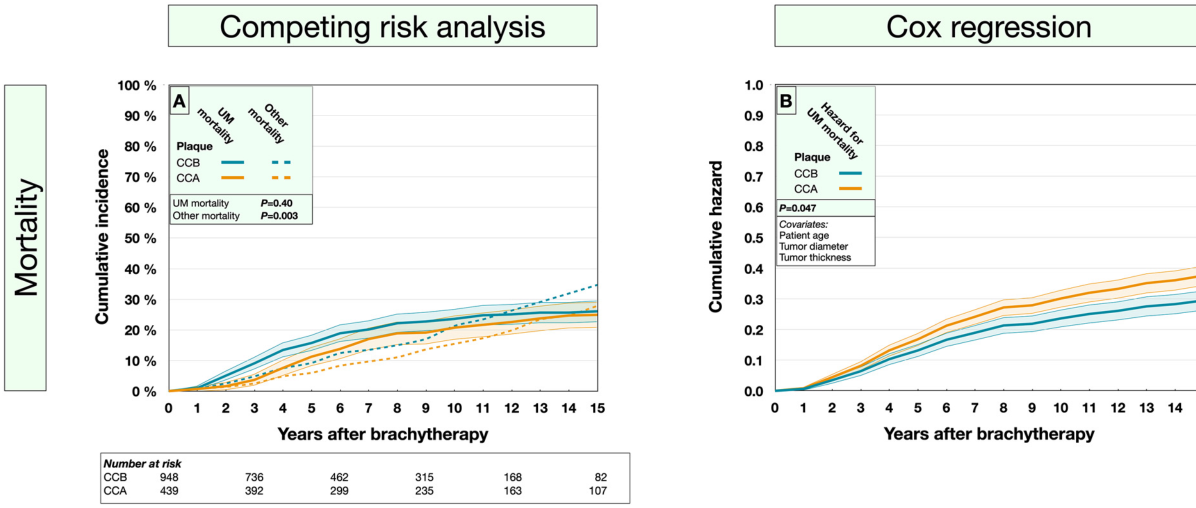

Fig. 2. Risk for mortality in patients treated with CCA versus CCB plaques. (A) Despite having tumors with greater diameter, thickness, and more advanced American Joint Committee on Cancer T-category, patients treated with CCB plaques had a similar incidence of uveal melanoma–related mortality as patients treated with CCA plaques. (B) When adjusting for patient age, tumor diameter, and tumor thickness, patients treated with CCA plaques had a greater cumulative Cox regression hazard for uveal melanoma–related mortality. Colored fields represent 95% CIs. *Abbreviation*: UM = uveal melanoma.

tumors and more advanced AJCC stages. In multivariate analysis with patient age, tumor diameter, and thickness as covariates, CCB plaques were associated with a 23% lower risk for uveal melanoma–related mortality (HR, 0.77; 95% CI, 0.59-0.99; Fig. 2B, Table E1).

Association between tumor recurrence and mortality

In patients who suffered from tumor recurrence, irrespective of what plaque type had been used, the tumors were diagnosed at a younger age and the patients had thicker tumors. Additionally, they had been treated with a smaller difference between plaque diameter and LBD, resulting in a higher cumulative incidence of uveal melanoma–related mortality (Fig. 3A,B).

The sample was categorized into 4 groups based on the difference between plaque diameter and LBD (0-3.9, 4.0-7.9, 8.0-11.9, and ≥12.0 mm). As the difference decreased, the cumulative incidence of tumor recurrences increased. In the group with the smallest difference (0-3.9 mm, corresponding to a safety margin of less than 2 mm), the 15-year incidence of tumor recurrence was 41% for CCA plaques and 39% for CCB plaques. Lower incidence rates were found for CCB plaques in categories with smaller difference compared with CCA plaques. For example, patients treated with CCA plaques in the 8.0-11.9-mm category had a 15-year incidence of tumor recurrences of 24%, which exceeded the 20% incidence for patients treated with CCB plaques in the 4.0-7.9-mm category (Fig. 3C,D).

In univariate Cox regressions with tumor recurrence as a time-varying covariate and in multivariate regressions with patient age, tumor diameter, and tumor thickness as covariates, tumor recurrence was associated with both uveal melanoma–related and all-cause mortality (Table 3).

Discussion

The main finding of this study is that ruthenium brachytherapy of uveal melanoma with 15-mm CCA plaques was associated with a greater risk for tumor recurrence and metastatic death than 20-mm CCB plaques.

A safety margin of at least 2 mm was planned to a similar extent for CCA and CCB plaques. However, on average, the difference between plaque diameter and LBD was greater with the CCB plaque. Interestingly, a smaller difference in plaque diameter minus LBD was as strong a risk factor for recurrence and mortality as tumor diameter. These outcomes were not improved by combining brachytherapy with TTT.

This author proposes that the increased recurrence rate observed after CCA brachytherapy may be attributed to the smaller difference in plaque and tumor diameter, which reduces tolerance for plaque misplacements and leads to subthreshold doses to areas of the tumor. In contrast, the average difference of 9.7 mm with CCB plaques ensures full doses to all tumor regions, even without optimal plaque placement. However, within the same categories of plaque diameter minus LBD, the recurrence rates remained higher after CCA brachytherapy. To achieve a somewhat comparable recurrence rate to that of a CCB plaque with a diameter 4.0- to 7.9-mm greater than the tumor (ie, LBD ≤ 16.2 mm), the CCA plaque required a diameter 8.0- to 11.9-mm greater than the tumor (ie, LBD ≤ 7.3 mm). This could indicate that a smaller difference between plaque and tumor diameter is not the sole reason for the inferior outcomes associated with CCA plaques. Their smaller diameter may pose a greater surgical challenge, especially for posteriorly located tumors that necessitate suturing the plaque at a significant distance from the limbus. This seems to be corroborated by our finding of a

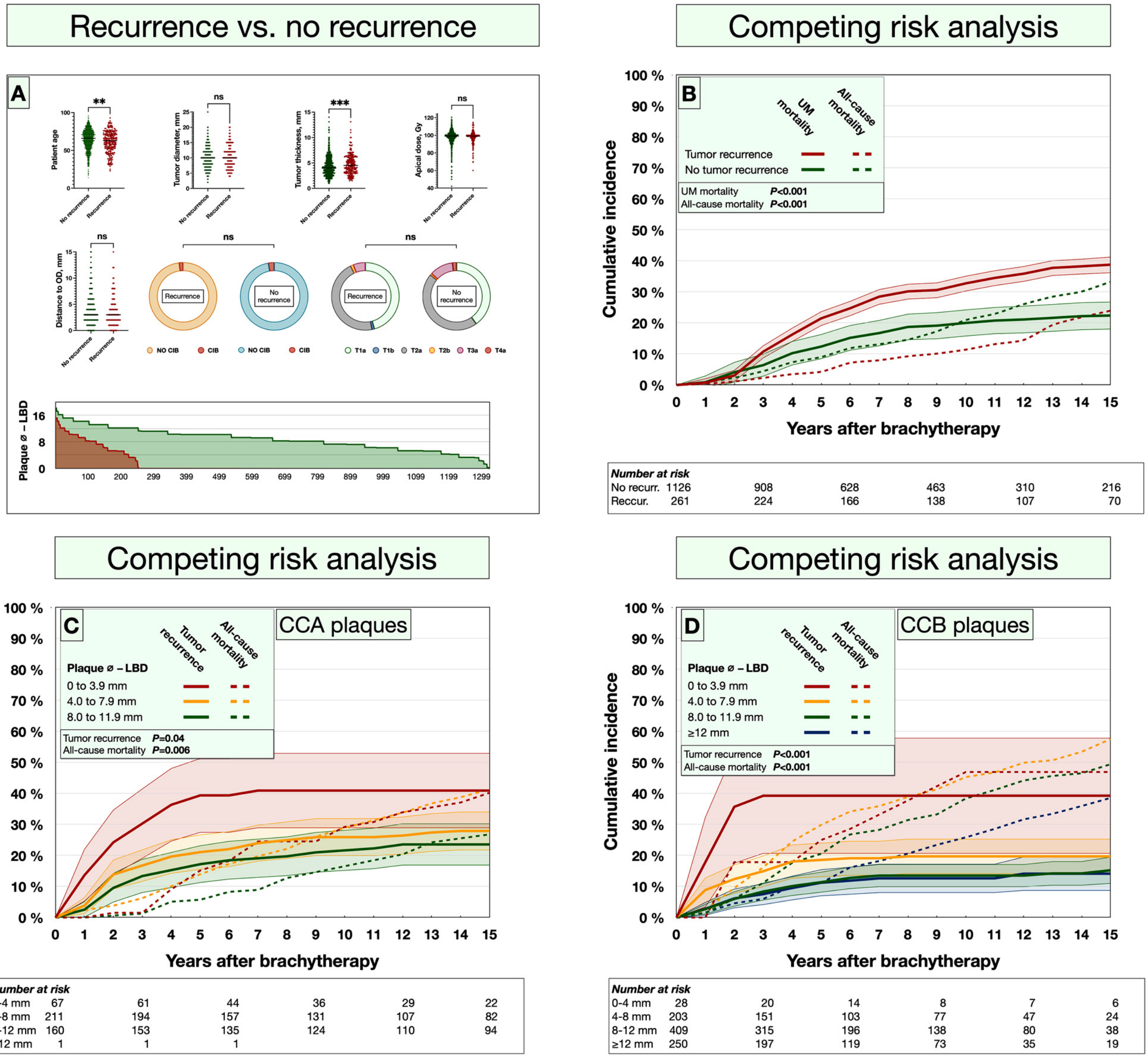

Fig. 3. Characteristics and survival of patients with tumor recurrence after plaque brachytherapy and cumulative incidence of tumor recurrence in relation to the difference between plaque diameter and LBD. (A) At the time of primary treatment, patients who suffered from tumor recurrence (red) were younger and had tumors with greater apical thickness. They had also been treated with smaller difference between plaque diameter and LBD (7.6 [standard deviation, 3.2] vs 9.0 [standard deviation, 3.3] mm; Mann-Whitney *U*, $P < .001$). (B) Patients who suffered from tumor recurrence (red) had a greater cumulative incidence of uveal melanoma−related mortality from the competing risk data. (C) The cumulative incidence of tumor recurrences increased with decreasing difference between CCA plaque and LBD, and was >40% 7 years after primary brachytherapy for the group, with a difference of 0 to 3.9 mm. (D) Similarly, the cumulative incidence of tumor recurrences increased with decreasing difference between CCB plaque and LBD, and was >39% 3 years after primary brachytherapy for the group, with a difference of 0 to 3.9 mm. For patients with a larger difference between plaque diameter and LBD however, incidence rates were lower in the CCB group. For example, patients with a plaque diameter minus LBD of 8.0 to 11.9 mm in the CCA group had a 15-year incidence of tumor recurrences of 24% (green line in C), which could be compared with the 20% incidence for patients treated, a smaller difference of 4.0 to 7.9 mm in the CCB group (yellow line in D). Colored fields represent 95% CIs. $^{**}P < .01$; $^{***}P < .001$. *Abbreviations*: LBD = largest basal tumor diameter; plaque ⌀ − LBD = plaque diameter minus LBD; UM = uveal melanoma.

trend for increasing recurrence-free survival proportion over time, which could suggest that there is a learning curve that gradually increases the likelihood of successful treatment over the course of several decades.

Plaque diameter minus LBD should not be confused with safety margin. The former represents the difference between plaque and tumor diameter, which will remain the same regardless of whether there is a 4-mm tumor-free margin on

Table 3 Cox regression with tumor recurrence as a time-varying covariate

Variate	B	S.E.	Wald	*P*	Exp(B)	95% CI lower	95% CI upper
Hazard for uveal melanoma–related mortality							
Univariate							
Tumor recurrence*	0.96	0.13	56.89	<.001§	2.60	2.03	3.33
Multivariate							
Tumor recurrence*	0.81	0.13	39.79	<.001§	2.26	1.75	2.91
Age at diagnosis†	0.05	0.13	0.14	.70	1.05	0.82	1.34
Tumor diameter, mm‡	0.11	0.02	36.07	<.001§	1.12	1.08	1.16
Tumor thickness, mm‡	0.08	0.03	6.20	.01§	1.09	1.02	1.16
Hazard for all-cause mortality							
Univariate							
Tumor recurrence*	0.37	0.09	16.22	<.001§	1.44	1.21	1.73
Multivariate							
Tumor recurrence*	0.27	0.09	8.33	.004§	1.30	1.09	1.56
Age at diagnosis†	0.92	0.08	134.66	<.001§	2.51	2.15	2.93
Tumor diameter, mm‡	0.08	0.01	41.43	<.001§	1.08	1.06	1.11
Tumor thickness, mm‡	0.05	0.02	4.61	.03§	1.05	1.00	1.10

Abbreviations: B = beta coefficient S.E. = standard error.
* Time-varying covariate.
† >70 vs ≤70 years.
‡ Per millimeter.
§ Significant *P* values.

1 side of the tumor and 2 mm on the other or a 6-mm tumor-free margin on 1 side and 0 mm on the other. On the other hand, the safety margin refers to the minimum tumor-free margin of plaque overlap outside the tumor in all directions. Importantly, safety margins should be viewed as an intention rather than a factual outcome. Even the slightest misplacement of the plaque will influence the actual margin, and it may not be possible to confirm if all areas of the tumor base have received tumoricidal radioactive doses until a chorioretinal atrophy has developed several months after treatment. Clearly, the risk for deviations from the planned safety margin and insufficient overlap is high when small plaques are used without intraoperative verification of their position. But even with indentation or right-angled transillumination of the sclera, correct placement within 1 or 2 mm cannot be guaranteed in all cases, and small plaque movements after attachment to the sclera may result from multiple factors, including a slight laxity of the plaque-holding sutures, small bleedings, tissue swelling or entrapment, plaque tilting, further surgical manipulations of the eye, or eye movements. Furthermore, the prescription isodose line does not necessarily align with the edge of the plaque. In treatment of thin tumors, the isodose line may lie inside the edge, whereas for thick tumors, it may be extended several millimeters in the lateral dimension.[14] Consequently, a standard safety margin of 2 mm may result in insufficient radiation doses to the tumor periphery, even with optimal placement of the plaque, particularly when thin tumors are treated without a minimum scleral dose.[38]

In 2005, Damato et al[25] reported 5-year actuarial recurrence rates of only 2% after plaque brachytherapy of 250 patients with CCA plaques and 208 patients with CCB plaques. This is dramatically lower than the incidence rates found herein. In the study by Damato et al, in contrast to our surgical technique, correct plaque positioning was confirmed by both indenting the sclera with a plastic template and by right-angled transillumination through drilled holes close to the posterior margin of the template. Further, a minimal scleral radiation dose of 300 to 400 Gy was used, and tumors were ineligible for plaque brachytherapy if their posterior tumor margin extended close to the optic disc.[25] The diameter of the optic nerve sheath is approximately 5 mm, and it has been shown that tumors with a posterior margin within 2 mm of the optic disc are at increased risk for treatment failure.[33,39,40]

Similarly, in a study from 2021, Jouhi et al[26] found no difference in the incidence of local recurrences after brachytherapy with 10- and 15-mm ruthenium plaques. However, even though plaque placements had been verified with indentation or transillumination with a vitrectomy probe at each plaque margin, eccentric plaque placement was associated with a higher HR for local recurrence compared with our findings (HR, 3.4 vs 1.6). Other important differences include an upper tumor diameter limit of 10 mm (whereas we applied no upper limit), a minimum scleral dose of 250 Gy (whereas we used no minimum dose), a cohort of 164 patients (whereas we analyzed 1387), and a slightly shorter follow-up (median, 8.1 vs 10.8 years).

Therefore, we conclude that the combination of smaller-diameter plaques and absence of verification of plaque placement leads to smaller differences between plaque diameter and LBD, less room and tolerance for inaccuracy in plaque placement, and an increased rate of tumor recurrence. In turn, this increases the risk for uveal melanoma—related mortality.

The link between CCA plaques, tumor recurrence, and death was consistent in Cox regression analyses and in cumulative incidence analyses from competing risk data. Further, our data corroborate previous observations that tumor recurrence is associated with an increased risk for uveal melanoma—related mortality. Vrabec et al[30] and Gragoudas et al[31] reported significantly shorter survival among patients who suffered from local tumor recurrences after cobalt-60 plaque brachytherapy and proton therapy, respectively. In 2016, an international collaboration reported a 10-year local recurrence rate of 11%, and that the group of patients who suffered from local tumor recurrence had shorter metastasis-free survival.[29]

Nevertheless, it has been debated whether the tumor recurrence is the cause of metastatic disease in itself or merely an indicator of increased metastatic potential from the onset.[29] If the latter was true, it would mean that there was a larger presence of aggressive features among the tumors that were treated with CCA plaques than among the significantly larger tumors treated with CCB plaques in the present study and that even though the smaller tumors had a greater metastatic potential, it was not reflected in the incidence of uveal melanoma—related mortality. This seems contradictory and is inconsistent with the findings herein, and this author believes that these results indicate that a local tumor recurrence may in itself increase the risk for metastasis. This view is corroborated by a recent study by Bagger et al,[41] who found that tumors that recur after plaque brachytherapy are not more likely to have abnormal chromosome 3 and 8q status at the baseline.

Our 5-, 10-, and 15-year local recurrence rates after ruthenium CCB plaque brachytherapy of 13%, 15%, and 15%, respectively, are similar to rates reported by other authors. In the large randomized Collaborative Ocular Melanoma Study, the 5-year secondary enucleation rate was 13% by the Kaplan-Meier method, and the 5-year treatment failure rate was 10%. The latter was defined as tumor growth by ≥15% in ultrasonography or a 250-μm expansion of tumor margins in photographs or clinical examination on 2 subsequent occasions.[42]

Each tumor recurrence brings significant distress to the patient, typically requiring repeat hospital visits and 1 or several treatments under general anesthesia. Patients who must undergo secondary enucleation may be particularly disheartened, as their primary treatment was intended to be eye-preserving. Additionally, treatment of recurrences adds a significant time and resource burden on the health care system.

The consequences of the increased risk for death are even more profound. Few patients will want to undergo plaque brachytherapy with a smaller plaque if they are informed that it can increase the risk for tumor recurrence and death, and few ocular oncologists will recommend it. In our own institution, we are now reducing the use of CCA plaques significantly and more frequently recommend enucleation of tumors in close proximity of the optic disc. We have also introduced a minimum scleral dose of 350 Gy for ruthenium brachytherapy and acrylic templates with perforations into which a right-angled transilluminator is introduced to enable ophthalmoscopic verification of accurate positioning of the template, and hence the plaque, with respect to the tumor.[25]

The findings presented herein do not contradict the current understanding of the metastatic process, in which micrometastases are seeded years before primary tumor diagnosis and treatment.[43-46] In the prognostic sense, secondary treatment of tumors that were initially small but have recurred or continued to grow should be likened to primary treatment of a large tumor. The prolonged tumor cell viability and additional mitoses that occur in a tumor that has not been fully sterilized by the first treatment likely increase the risk that aggressive traits such as monosomy 3 and *BAP1* mutations will develop, as is true for any growing uveal melanoma. This suggests that tumor cells that enter the systemic circulation in later stages of the disease have a better ability to establish viable micrometastases. This is further corroborated by one of our recent publications, in which we show that patients with untreated primary uveal melanoma have a very high risk for metastasis.[47] If metastatic spread always occurs before primary tumor treatment, then such treatment would not influence survival at all.

The main strengths of this study are the large cohort, the long and complete follow-up, the cross-referencing between several data sources, the mandatory medical death certificates, and high autopsy rates. In the 1980s, >80% of all deceased Swedish inhabitants underwent autopsy. In the 1990s, this fell to 34%, followed by even lower rates in recent years.[48] For patients dying from a diagnosed or suspected cancer, autopsy rates have remained relatively high, especially when the cause of death cannot be reliably determined by other means, for example, radiologic examinations, biopsies, or laboratory tests before death.[49] Because misclassifications still occur, most often in this cohort from metastatic uveal melanoma being erroneously recorded as metastatic cutaneous melanoma, the registered cause of death was also cross-checked with medical records.

Limitations of this study include its retrospective nature, for which we compensated with analyses of all major confounders for tumor recurrence. These variates play a crucial role that likely reduces the scope for a randomized trial: CCA and CCB plaques are not intended to be used for tumors of the same size. Further, even though treatments were administered over a time span of 4 decades by more than a dozen ocular oncologists, they were all performed at the same center, with routines and practices that may not necessarily be identical to other institutions. The inferior oblique muscle had not always been removed when plaques were placed inferotemporally, and in such cases, the resulting extra distance between the plaque and tumor was not accounted for. This could have led to insufficient radioactive doses but should have a similar influence on CCA and CCB plaques. Last, our analyses do not include genetic or cytogenetic features that are strongly associated with the risk for metastatic death.[45,50,51] However, these factors become more prevalent with increasing tumor size, which would consequently have biased the results toward better survival after brachytherapy with CCA plaques.[45,52,53]

Conclusion

Compared with standard 20-mm ruthenium CCB plaques, brachytherapy with 15-mm CCA plaques is associated with a higher risk for tumor recurrence. Adjusted for tumor size, CCA plaques are also associated with uveal melanoma −related mortality. The risk for recurrence or uveal melanoma-related mortality was not lower for patients treated with adjunct TTT. Patient survival can likely improve if local treatment failures are avoided by methods that increase the accuracy of plaque placements and by wider treatment margins. One of these 2 should not be used to justify absence of the other. Advocating for large differences in plaque and tumor diameter to compensate for careless plaque placement is not advisable, and, as shown by others, measures that improve placement accuracy cannot compensate for inadequate treatment margins. Previous literature has shown that centers that verify plaque positioning by scleral indentation or transillumination, use minimum scleral doses, and consider enucleation for juxtapapillary tumors have achieved lower recurrence rates. It is encouraged for other ocular oncology centers to evaluate their outcomes with small-diameter plaques. If not already implemented, they should consider increasing safety margins and adapting effective methods to verify accurate plaque positioning. As this would make a smaller number of tumors eligible for treatment with CCA plaques, their use will likely be reduced.

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